Summary :

1. Convert leg fram into body frame:

for $\vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, convert to $\vec{P}' = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$.

$$T(\text{rear-left}) = \begin{bmatrix} -1 & 0 & 0 & \frac{-X_0}{2} \\ 0 & 1 & 0 & \frac{Y_0}{2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$T(\text{rear-right}) = \begin{bmatrix} -1 & 0 & 0 & \frac{-X_0}{2} \\ 0 & -1 & 0 & \frac{-Y_0}{2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$T(\text{front-left}) = \begin{bmatrix} 1 & 0 & 0 & \frac{X_0}{2} \\ 0 & 1 & 0 & \frac{Y_0}{2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T(\text{front-right}) = \begin{bmatrix} 1 & 0 & 0 & \frac{X_0}{2} \\ 0 & -1 & 0 & \frac{-Y_0}{2} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

$$\vec{P}'_{(\text{body fram})} = T \cdot \vec{P}' \Rightarrow \vec{P}_{3\times 1}$$

X_0 = body length from leg frame

Y_0 = body width from leg frame.

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2. IMU Static Body Posture Control:

for $\Delta R \; \Delta\theta_B = \begin{bmatrix} \theta_{roll} \\ \theta_{pitch} \\ 0 \end{bmatrix}$, $\vec{P}_{body} = \begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = (T \cdot \vec{P}')_{3\times 1}$.

$$\Delta\vec{P}_B = \theta_P \begin{bmatrix} z_B \\ 0 \\ -x_B \end{bmatrix} + \theta_r \begin{bmatrix} 0 \\ -z_B \\ y_B \end{bmatrix},$$

$$\boxed{\Delta\vec{P}_{3\times 1} = T^{-1} \cdot \Delta\vec{P}_B}$$

3. Convert foot displacement into joint displacement.

$$\dot{P} = \frac{\Delta \vec{P}}{dt} \quad , \quad dt = \frac{1}{\text{loop rate (Hz)}}$$

$$\dot{q} = J^{\#} \cdot \dot{P} \quad \rightarrow \quad \Delta q = \int \dot{q} \cdot dt = \int J^{\#} \cdot \frac{\Delta \vec{P}}{dt} \cdot dt = J^{\#} \cdot \Delta \vec{P}$$

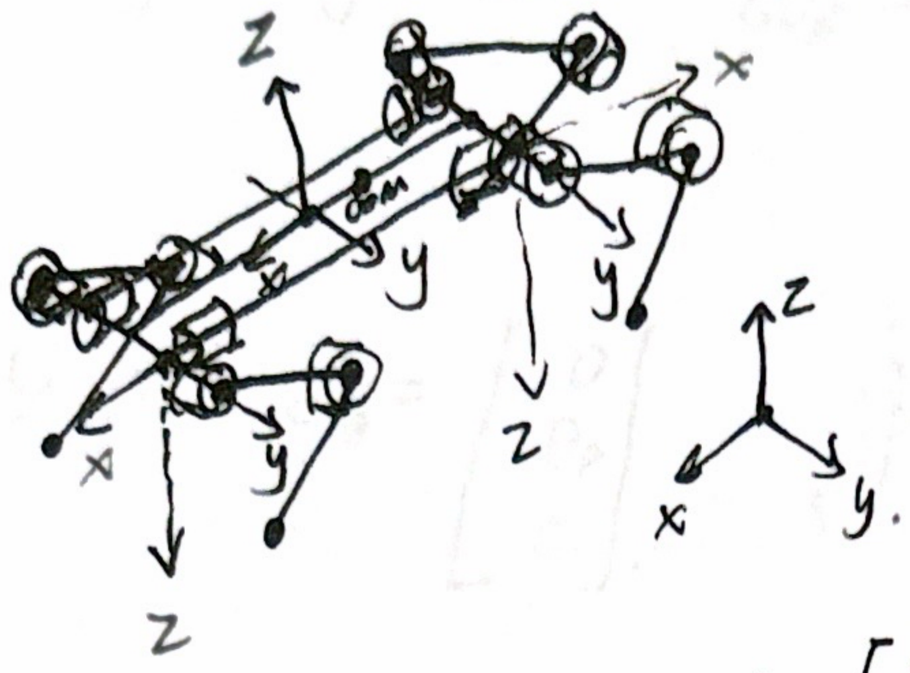
Note: for $\Delta \vec{P} = \theta_P \cdot \begin{bmatrix} Z_B \\ 0 \\ -X_B \end{bmatrix} + \theta_r \begin{bmatrix} 0 \\ -Z_B \\ Y_B \end{bmatrix}$,

Converging radius: $|\theta| < \frac{\pi}{2}$.

error $\approx \frac{\theta^3}{3}$, so if large θ, then

$$\Delta \vec{P} = \tan(\theta_P) \begin{bmatrix} Z_B \\ 0 \\ -X_B \end{bmatrix} + \tan(\theta_r) \begin{bmatrix} 0 \\ -Z_B \\ Y_B \end{bmatrix}.$$

Can calculate error clamping by let Error(θ) < 0.1 rad.



front to rear leg frame length: X_0

left to right leg frame length: Y_0

⋮

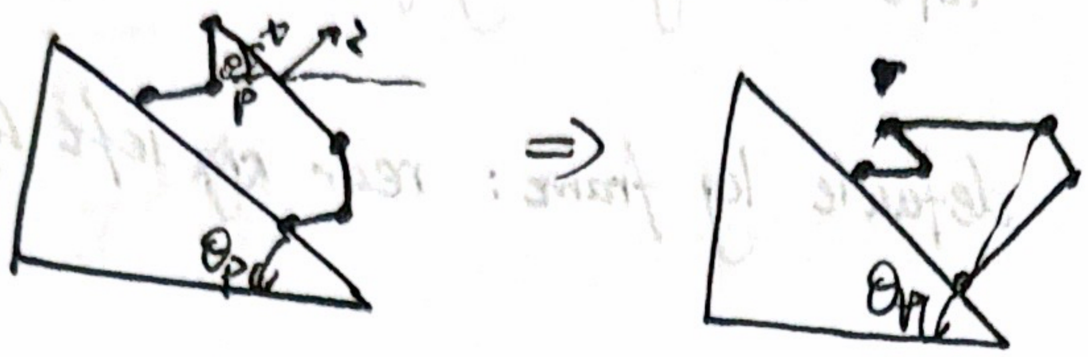
default leg frame: rear left leg.

for rear left leg: $\vec{P}_L = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in leg frame:

in body frame: $\vec{P}_B = \begin{bmatrix} -x - \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix}$

$$\begin{cases} \vec{P}_{B_{rL}} = \begin{bmatrix} -x - \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix} & \text{for rear left leg.} \\ \vec{P}_{B_{rr}} = \begin{bmatrix} -x - \frac{X_0}{2} \\ -y - \frac{Y_0}{2} \\ -z \end{bmatrix} & \text{for rear right leg.} \\ \vec{P}_{B_{fL}} = \begin{bmatrix} x + \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix} & \text{leg position in body frame.} \\ \vec{P}_{B_{fr}} = \begin{bmatrix} x + \frac{X_0}{2} \\ -y - \frac{Y_0}{2} \\ -z \end{bmatrix} & \text{for front right leg.} \end{cases}$$

Pitch:



$\Rightarrow$





$\Rightarrow$

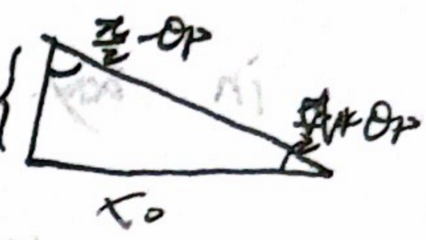


$\frac{\pi}{2} - \frac{\pi}{2} + \theta_p = \theta_p$

$\rightarrow$



$\Rightarrow$



$$\frac{h}{x_0} = \tan(\frac{\pi}{2} + \theta_p) \rightarrow h = x_0 \cdot \tan(\frac{\pi}{2} + \theta_p)$$

So if commading front leg with height H_0.

where in leg frame: $\vec{P} = \begin{bmatrix} x \\ 84.26 \\ H_0 \end{bmatrix}$.

then rear leg: $\vec{P} = \begin{bmatrix} 0 \\ 84.26 \\ H_0 + h \end{bmatrix}$, $h = x_0 \tan(\frac{\pi}{2} + \theta_p)$

So for pitch correction: θ_p = Pitch error

then for normal standing: we have original neutral standing:

$\vec{P}_{L_0} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in leg frame,

then front or real leg correction: $\vec{P}'_{0_L} = \begin{bmatrix} x \\ y \\ z \pm \frac{h}{2} \end{bmatrix}$ $h = x_0 \tan(\theta_p)$.

then $\vec{q}_0 = J^{\#} \cdot \vec{P}_{0_L}$ $\vec{q} = \vec{q}_0 + \dot{q}_0 \cdot \Delta t$.

$\dot{q}_0 \Delta t = J^{\#} \cdot \vec{P}'_L$

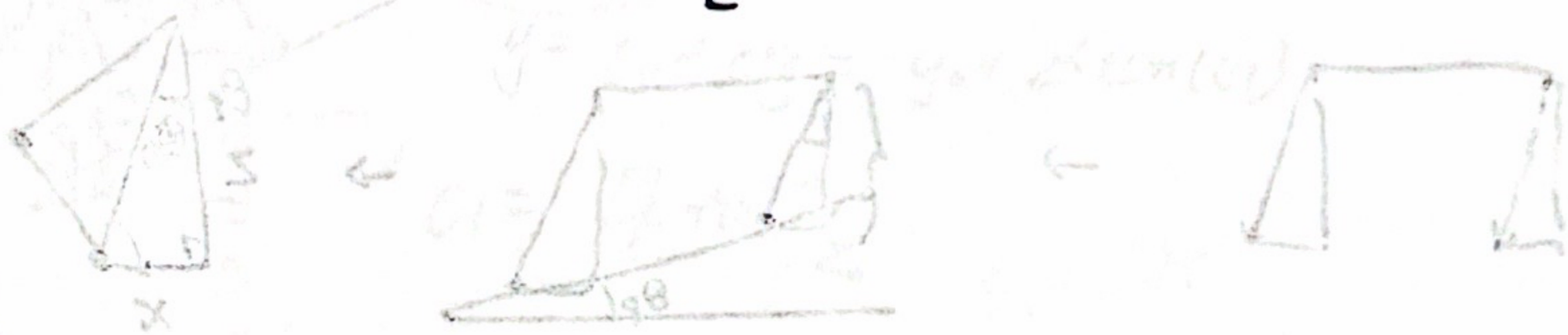
correcton angles $\Delta \vec{P} = J^{\#} \cdot \dot{P}'_L \cdot \Delta t$

tilting angle vector: $\Delta\theta_B = \begin{bmatrix} \theta_{roll} \\ \theta_{pitch} \\ \theta_{yaw} \end{bmatrix}$.

if only pitch: $\Delta\theta_B = \begin{bmatrix} 0 \\ \theta_p \\ 0 \end{bmatrix}$ $\vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in leg frame;

$$\Delta\vec{p} = \Delta\theta_B \times \vec{P} = \begin{bmatrix} i & j & k \\ 0 & \theta_p & 0 \\ x & y & z \end{bmatrix} = \hat{i}\cdot(z\theta_p) - \hat{j}\cdot 0 + \hat{k}\cdot(-x\theta_p).$$

$$\Delta\vec{p} = \begin{bmatrix} z\theta_p \\ 0 \\ -x\theta_p \end{bmatrix}. \quad \text{(front leg.)}$$



Taylor: $\sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!}\cdot(x-a)^n$ for $\tan(\theta)$.

as $\theta \to 0$.

$n=0$: $\frac{\tan(0)}{1}\cdot(\theta-0)^0 = 0$ $\quad \sec(0) = \frac{1}{\cos(0)} = 1$.

$n=1$: $\frac{\sec^2(0)}{1!}(\theta-0)^1 = \theta$. $\quad \left((\cos(\theta))^{-1}\right)'$

$n=2$: $2\sec(\theta)\cdot 0$ $\quad = (-1)(\cos(\theta))^{-2}\cdot -\sin\theta$

$n=3$: $\frac{1}{3!}\cdot(\theta)^3 = \frac{1}{6}\theta^3$. $\quad = \frac{\sin\theta}{(\cos(\theta))^2}$

$\frac{1}{3}\theta^3$.

$\tan(\theta) = \theta + \frac{1}{6}\theta^3 + \cdots$ $\quad \left(\sin\theta\cdot(\cos(\theta))^{-2}\right)'$

when $\theta \to 0$. converging radius: $|\theta| < \frac{\pi}{2}$ $\quad \cos\theta(\cos(\theta))^{-2} + \sin\theta(-2)(-1)\cdot(\cos\theta)^{-3}\sin\theta$

$\tan(\theta) \approx \theta$. around $\theta = 0$. $\quad = 1$

error in $\tan(\theta)$ if $\tan(\theta) \approx \theta$.

$$\text{error} = \frac{\theta^3}{3} + \frac{2\theta^5}{15} + \frac{17\,\theta^7}{315} \cdots$$

for $\theta = \frac{\pi}{3}$ (60°): error = 0.38 rad $\approx$ 22 deg.

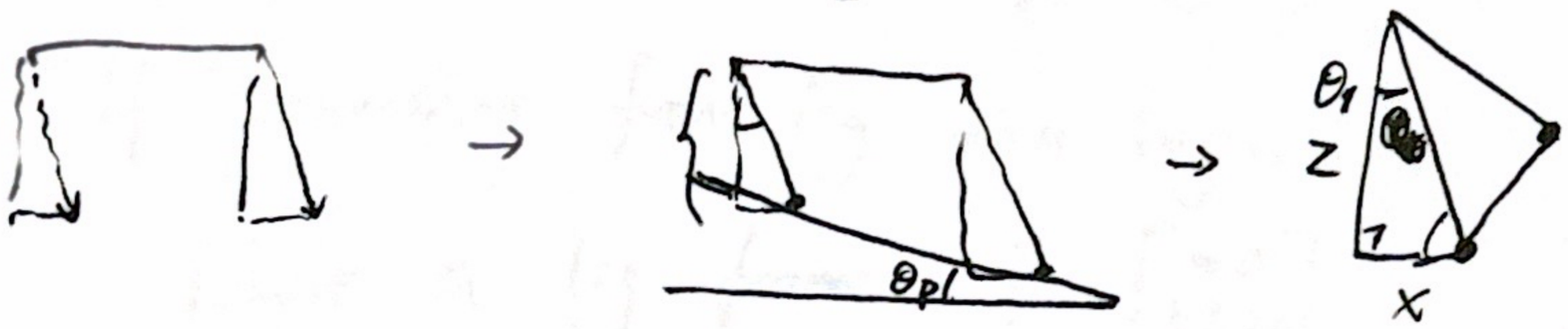
if wanna control error within 1 deg: $\frac{\theta^3}{3} = 1$

$\theta = \sqrt{3} = 1.44$ rads. deg = 0.025 rad.

usually normal tilting angle $< \frac{\pi}{4}$ (45°).

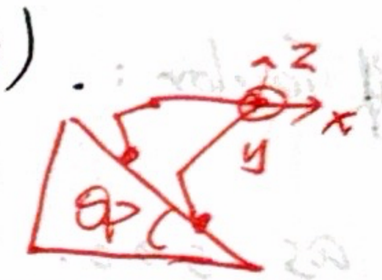
for $\theta = \frac{\pi}{4}$: error = 0.16 rad = 9.25 deg. ×

for $\theta = \frac{\pi}{6}$: error = 0.0478 rad = 2.74 deg ✓



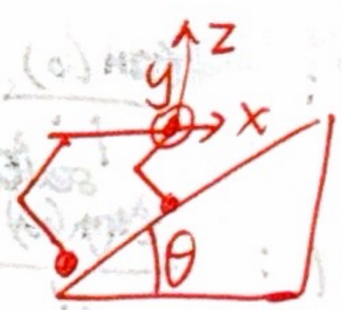
$\theta_1 = \theta_P$.

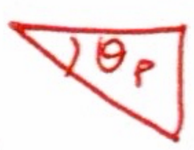
$\theta_2 = \tan^{-1}\left(\frac{x}{z}\right)$ $\theta_2 = \frac{\pi}{2} - \theta_1$ $x = z \cdot \tan(\theta_1)$.



so when $z = z_0 + h$

$h = x_0 \tan(\theta_{pitch})$





then $x = x_0 + l$

$l = z_0 \tan(\theta_{pitch}) \tan(\theta_1) = z' \tan(\theta_1)$.

$\theta_1 = \tan^{-1}\left(\frac{x}{z}\right)$. $\tan\left(\tan^{-1}\frac{x}{z}\right) = \frac{x}{z}$

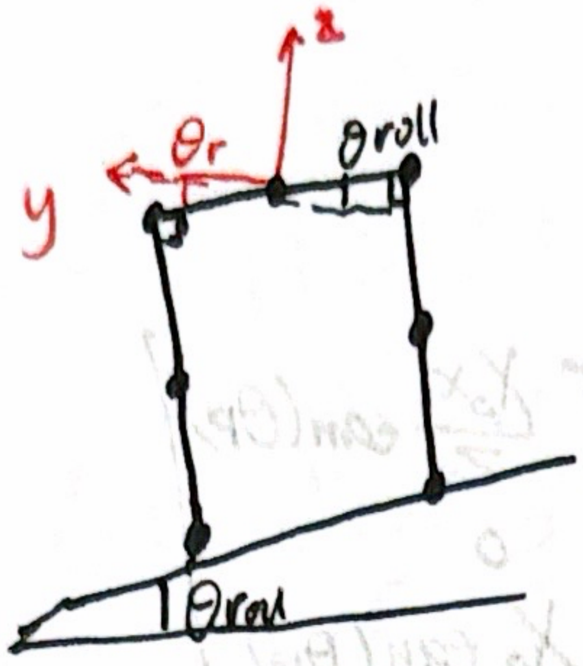
for pitch error θ_p.

$$\begin{cases} z = z_0 + x_0 \tan(\theta_{pitch}) \\ x = x_0 + (z_0 + x_0 \tan(\theta_{pitch})) \tan\left(\frac{x_0}{z_0}\right) \end{cases}$$

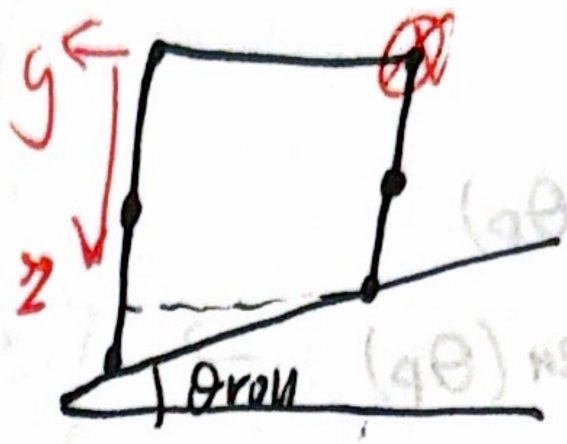
x_0: body length

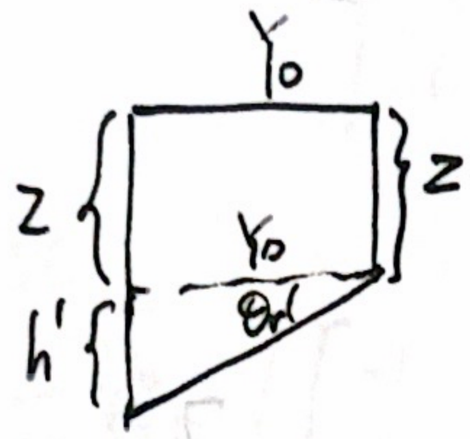
$\frac{x_0}{z_0}$: commanded neutral pose original.

roll :



$\Rightarrow$

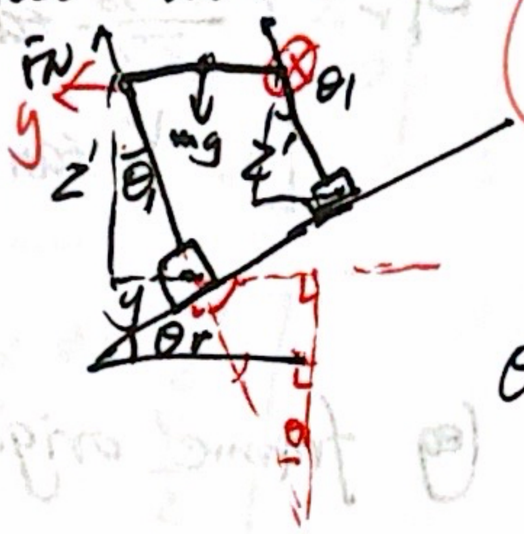




$$\begin{cases} Y_0 \tan(\theta_r) = h' \\ z' = z_0 + h' \end{cases}$$

foot paw thickness w_0 :





$$\Delta y = z \cdot \tan(\theta_r) = z' \cdot \tan(\theta_1)$$

$$y = y_0 + \Delta y = y_0 + z' \tan(\theta_1)$$

$$\theta_1 = \tan^{-1}\frac{y}{z_0}$$

$\theta_1 = \theta_r$

$$y = y_0 + z' \tan(\theta_1)$$

$$y = y_0 + \left((z_0 + Y_0 \tan(\theta_r)) \right) \cdot \frac{y_0}{z_0}$$

so

$$\begin{cases} z' = z + Y_0 \tan(\theta_r) \\ y = y_0 + (z_0 + Y_0 \tan(\theta_r))\left(\frac{y_0}{z_0}\right) \end{cases}$$

$$y = 2y_0 + \frac{Y_0 y_0}{z_0} \tan(\theta_r)$$

Summary :

for pitch error θ_{pitch} :

$$\begin{cases} z = z_0 + X_0 \tan(\theta_P) \\ x = 2x_0 + \dfrac{X_0 x_0}{z_0} \tan(\theta_P) \\ y = y_0 \end{cases} \Rightarrow \Delta P_p = \begin{bmatrix} \dfrac{X_0 x_0}{z_0} \tan(\theta_P) \\ 0 \\ X_0 \tan(\theta_P) \end{bmatrix}$$

for roll error θ_{roll} :

$$\begin{cases} z = z_0 + Y_0 \tan(\theta_{roll}) \\ x = x_0 \\ y = 2y_0 + \dfrac{Y_0 y_0}{z_0} \tan(\theta_{roll}) \end{cases} \Rightarrow \Delta P_r = \begin{bmatrix} 0 \\ \dfrac{Y_0 y_0}{z_0} \tan(\theta_{roll}) \\ Y_0 \tan(\theta_{roll}) \end{bmatrix}.$$

X_0 : body length from front leg frame origin to rear leg frame origin.

Y_0 : body width from left leg frame origin to right leg frame origin.

$$\begin{bmatrix} X_0 \\ y_0 \\ z_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 84.26 \\ 380.0 \end{bmatrix} mm = \text{neutral standing position in leg frame.}$$

Convert leg frame to body frame:

$$\vec{P}_{BrL} = \begin{bmatrix} -x - \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix} \qquad \vec{P}_{BfL} = \begin{bmatrix} x + \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix}$$

$$\vec{P}_{BrR} = \begin{bmatrix} -x - \frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \end{bmatrix} \qquad \vec{P}_{BfL} = \begin{bmatrix} x + \frac{X_0}{2} \\ -y - \frac{Y_0}{2} \\ -z \end{bmatrix}$$

mo geneous coordinate :

use : $\vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$.

then $T = \begin{bmatrix} -1 & 0 & 0 & -\frac{X_0}{2} \\ 0 & 1 & 0 & \frac{Y_0}{2} \\ 0 & 0 & 1 & z_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

test :

$T \cdot \vec{P} = \begin{bmatrix} -1 & 0 & 0 & -\frac{X_0}{2} \\ 0 & 1 & 0 & \frac{Y_0}{2} \\ 0 & 0 & -1 & -0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -x + -\frac{X_0}{2} \\ y + \frac{Y_0}{2} \\ -z \\ 1 \end{bmatrix}$ ✓

for body posture error $= \begin{bmatrix} \theta_r \\ \theta_p \\ 0 \end{bmatrix}$. (we don't care yaw yet).

$\Delta\theta_B = \begin{bmatrix} \theta_r \\ \theta_r \\ 0 \end{bmatrix}$. $\Delta\vec{P}_B = \begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix}$ in Body frame

$\Delta\theta_B \times \vec{P} = \begin{bmatrix} i & j & k \\ \theta_r & \theta_p & 0 \\ x_B & y_B & z_B \end{bmatrix} = \hat{i}(\theta_p z_B) - \hat{j}(\theta_r z_B) + \hat{k}(\theta_r y_B - x_B \theta_p)$

$\Delta\vec{P}_B = \begin{bmatrix} \theta_p z_B \\ -\theta_r z_B \\ \theta_r y_B - x_B \theta_p \end{bmatrix} = \theta_p \cdot \begin{bmatrix} z_B \\ 0 \\ -x_B \end{bmatrix} + \theta_r \begin{bmatrix} 0 \\ -z_B \\ y_B \end{bmatrix}$. (rear. front (e convention).